

VISUAL TOUR

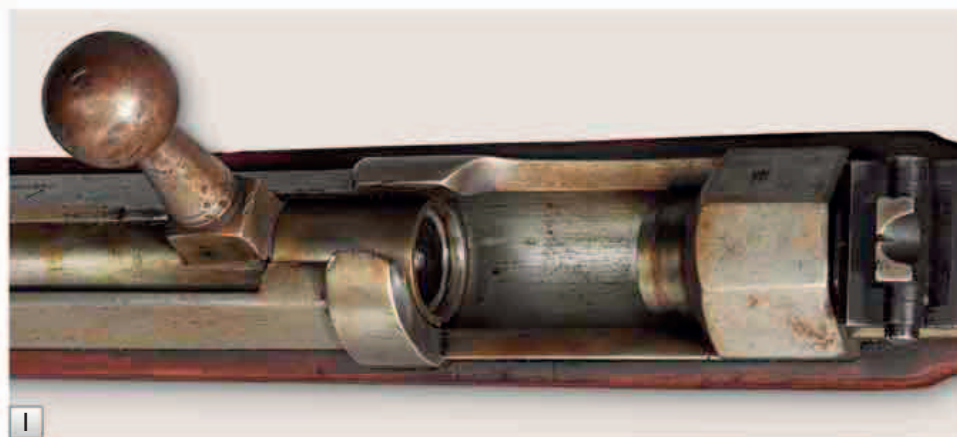
DREYSE NEEDLE-FIRE RIFLE

German gunsmith **Johann Nikolaus von Dreyse** invented the first rotating bolt for loading a rifle at the breech (see p.304). It sealed the breech much more securely than previous breech-loaders did and ensured that the energy of the expanding gas propelled the bullet forward. The rifle was also revolutionary in using a long, thin firing pin to pierce a “self-consuming” paper cartridge, both drawn from the designs of Jean Samuel Pauly, Dreyse’s employer.



► BOLT AT REAR (BREECH OPEN)

Bolt action provides the rifle with an effective opening breech mechanism. The bolt was connected to a needle-shaped firing pin (opposite). Before the bolt could be unlocked, the firing pin would be retracted using the catch at the rear of the bolt. The bolt would then be rotated using the handle and pulled rearward, opening the breech. Once the breech was open, a cartridge was placed into it to load the gun.



► BOLT AT FRONT (BREECH CLOSED)

The bolt was closed by pushing the handle forward and rotating it. Doing this sealed the breech and also cocked the gun ready for firing. This gun fired paper cartridges that were not only self-contained (containing primer, charge, and bullet) but also “self-consuming.” The cartridges would combust fully, leaving behind no shell or residue to eject, allowing the weapon to be reloaded very efficiently.

